
GENERAL RELATIVITY TUTORIAL

Problem set 2, 20.09.2019.

■ PROBLEM 5 Constant acceleration, part I.

Consider the equation of motion in SR for a point particle

$$\frac{d\vec{p}}{dt} = \vec{F} , \quad \vec{p} = m\gamma\vec{v} , \quad \gamma = \frac{1}{\sqrt{1-\beta^2}} , \quad \beta = \frac{v}{c} . \quad (5.1)$$

Consider the situation where the force $\vec{F}$ points along the x -axis and has the constant value $F = mg$. Assume that the velocity and position are zero at the time $t = 0$.

(a) Show that the motion of the particle is hyperbolic,

$$\left(x + \frac{c^2}{g}\right)^2 - (x^0)^2 = \left(\frac{c^2}{g}\right)^2 , \quad (5.2)$$

where $x^0 = ct$ and $y = z = 0$.

(b) Let $\mathbf{I}$ denote the inertial system where the particle is at rest at $t = 0$. Show that the proper time (times c) of a clock following the accelerated particle is given by

$$\tau = \frac{c^2}{g} \operatorname{arcsinh} \left(\frac{gx^0}{c^2} \right) , \quad (5.3)$$

i.e.

$$x^0 = \frac{c^2}{g} \sinh \left(\frac{g\tau}{c^2} \right) . \quad (5.4)$$

(c) Show that the transformation from an inertial system $\mathbf{I}'$, where the accelerated particle is at rest at proper time τ at $x' = 0 = x^{0'}$ (and $y' = z' = 0$), to $\mathbf{I}$ is given by

$$x = \frac{c^2}{g} \left[\cosh \left(\frac{g\tau}{c^2} \right) - 1 \right] + x' \cosh \left(\frac{g\tau}{c^2} \right) + x^{0'} \sinh \left(\frac{g\tau}{c^2} \right) , \quad (5.5)$$

$$x^0 = \frac{c^2}{g} \sinh \left(\frac{g\tau}{c^2} \right) + x' \sinh \left(\frac{g\tau}{c^2} \right) + x^{0'} \cosh \left(\frac{g\tau}{c^2} \right) . \quad (5.6)$$

■ PROBLEM 6 Rotating coordinate system.

In order to find the line element of a uniformly rotating reference frame, we can start from flat space in cylindrical coordinates,

$$ds'^2 = g'_{\mu\nu} dx'^{\mu} dx'^{\nu} = -c^2 dt'^2 + d\rho'^2 + dz'^2 + \rho'^2 d\varphi'^2 , \quad (6.1)$$

and perform the following spatial transformation for axis of rotation z ,

$$\rho = \rho' , \quad z = z' , \quad \varphi = \varphi' + \omega t , \quad (6.2)$$

where ω is the constant angular velocity of rotation.

- (a) What is the line element ds^2 of the rotating coordinate system?
- (b) What would be the circumference of a circle in terms of the coordinate ρ as measured by an observer in a rotating frame? Discuss the physical meaning of the case when $\rho \geq \frac{c}{\omega}$.

Hint: An observer on a given space-time measures the time interval between two events as $d\tau^2 = -g_{00}dt^2$, and the space interval as $dl^2 = \left(g_{ij} - \frac{g_{0i}g_{0j}}{g_{00}}\right) dx^i dx^j$.